AWS SAA-C03 Study Guide

Module 5
Gap-Closing Supplement

Aligned to the official SAA-C03 exam guide • In-scope services not yet covered • Thinly covered task-statement concepts • Out-of-scope services to skip • Naming updates

1. Why This Module Exists

The official exam guide publishes an explicit in-scope service list plus task statements per domain. Modules 1–4 and the cheat sheet cover every task statement, but 14 in-scope services received no coverage and five concepts were treated thinly. This module closes those gaps so nothing on the official list is unfamiliar on exam day. These services appear mostly as distractors or single-question one-liners; recognize them, know their one sentence, and move on.

2. In-Scope Services Not Previously Covered

2.1 Networking

Service	What it is	Exam trigger
AWS Client VPN	Managed OpenVPN-based endpoint letting individual users (laptops) connect securely into a VPC over TLS; authenticates via AD, mutual certs, or SAML/IAM Identity Center. Contrast: Site-to-Site VPN connects NETWORKS (office ↔ VPC); Client VPN connects PEOPLE (remote workers ↔ VPC). Split-tunnel option sends only VPC traffic through the tunnel	'Remote employees need secure access to private VPC resources from anywhere' → Client VPN. 'Connect the data center' → Site-to-Site VPN
AWS Resource Access Manager (RAM)	Share AWS resources across accounts/OUs in an Organization without duplicating them: subnets (VPC sharing), Transit Gateways, Route 53 Resolver rules, License Manager configs, Capacity Reservations, Outposts. The receiving account uses the resource as if its own	'Multiple accounts must launch resources into the SAME VPC/subnets' → VPC sharing via RAM. 'Share the Transit Gateway with all accounts' → RAM

2.2 Management & governance

Service	What it is	Exam trigger
AWS Service Catalog	Curated catalog of approved CloudFormation-based products that end users self-provision; admins control versions, constraints (allowed instance sizes, regions), and launch roles; users don't need IAM permissions on the underlying resources	'Let developers deploy pre-approved architectures without giving them broad IAM rights' / 'enforce standardized, compliant provisioning'
AWS License Manager	Track and enforce software license usage (BYOL cores/sockets/vCPUs) across accounts; ties to Dedicated Hosts for host-affinity licenses; can block launches that would breach limits	'Track Windows/SQL Server/Oracle licenses', 'prevent exceeding license entitlements'
AWS Well-Architected Tool	Free console tool: answer questions per pillar about a workload, get risks (HRIs) and improvement plans against the Well-Architected Framework	'Review a workload against AWS best practices' / 'document architectural risks'
Amazon Managed Grafana	Managed Grafana dashboards querying CloudWatch, Prometheus, X-Ray, OpenSearch, Timestream etc.; SSO via Identity Center	'Existing Grafana dashboards / visualize metrics from multiple sources without managing servers'
Amazon Managed Service for Prometheus	Managed, scalable Prometheus-compatible metrics store, typically for EKS/container metrics; PromQL; pairs with Managed Grafana	'Kubernetes metrics with Prometheus, no self-managed server'

2.3 Containers & compute

Service	What it is	Exam trigger
Amazon ECS Anywhere	Run ECS tasks on YOUR on-prem servers/VMs (registered as external instances via SSM agent) while the ECS control plane stays in AWS	'Use the same ECS tooling for containers in the data center'
Amazon EKS Anywhere	Deploy and operate Kubernetes clusters on-prem (bare metal/VMware) using EKS-consistent tooling; AWS support available	'Run Kubernetes on-prem with AWS-supported distribution and tooling'
Amazon EKS Distro	The open-source Kubernetes distribution EKS itself uses; run it anywhere yourself	'Same K8s version/patches as EKS, self-managed anywhere'
VMware Cloud on AWS	VMware's SDDC stack (vSphere, vSAN, NSX) running on dedicated AWS bare-metal hosts; migrate VMs with vMotion/HCX with no re-platforming; integrates with native AWS services	'Migrate thousands of VMware VMs quickly without changing tooling or converting them'. (Rehost/relocate strategy)
AWS Serverless Application Repository	Marketplace of deployable serverless apps/components packaged with SAM; publish privately or publicly	'Reuse/share serverless application templates across teams'

2.4 Integration, analytics, media, mobile

Service	What it is	Exam trigger
Amazon AppFlow	No-code managed integration moving data between SaaS apps (Salesforce, ServiceNow, Slack, Zendesk, SAP) and AWS services (S3, Redshift), on schedule or event; field mapping, filtering, encryption in flight; PrivateLink-capable	'Ingest Salesforce/SaaS data into S3 or Redshift without writing integration code'
AWS Data Exchange	Find, subscribe to, and use third-party datasets (financial, healthcare, weather) delivered into S3 or via API; providers publish data products	'Consume licensed third-party datasets for analytics'
Amazon Kinesis Video Streams	Ingest, store, and process video/audio streams from cameras and devices; integrates with Rekognition Video and SageMaker; WebRTC for two-way real-time	'Stream video from IoT cameras for ML analysis'
Amazon Elastic Transcoder	Legacy managed media transcoding (convert videos to formats/resolutions for devices) from S3 to S3. AWS now steers new work to MediaConvert (out of scope), but Transcoder remains in scope	'Convert uploaded videos into multiple formats for mobile/web playback'
AWS Device Farm	Test mobile and web apps on real physical devices (iOS/Android/browsers) hosted by AWS; automated and remote manual testing	'Test the app on many real phone models without buying devices'

EXAM TIP: Most of these show up as distractors. The skill is recognizing what each one is NOT: Client VPN is not Site-to-Site; RAM shares resources, it doesn't move them; Service Catalog provisions approved stacks, it isn't a config-compliance tool (that's Config); Elastic Transcoder converts video, it doesn't stream it.

3. Task-Statement Concepts Covered Thinly Before

3.1 Immutable infrastructure (Domain 2, Task 2.2)

Mutable infrastructure is patched, updated, and reconfigured in place over time — servers drift, and 'it works on server 3 but not server 7' becomes normal. **Immutable** infrastructure is never modified after deployment: to change anything, you build a new artifact (a new AMI, container image, or Lambda version) and replace the running fleet, then discard the old one. Benefits: consistent, reproducible, easy rollback (redeploy the previous artifact), no configuration drift, and it forces statelessness.

- AWS building blocks: golden AMIs via EC2 Image Builder → launch template new version → ASG instance refresh (rolling replacement); container images in ECR → new ECS task definition revision; CloudFormation for reproducible stacks; blue/green deployments via ALB target groups or Route 53 weighted records.

EXAM TIP: 'Ensure all instances are identical / eliminate configuration drift / rapid reliable rollback' → immutable infrastructure: bake an AMI, deploy via launch template + instance refresh; never SSH in to patch.

3.2 Service quotas & throttling for standby environments (Task 2.2)

Service quotas (limits) are PER ACCOUNT PER REGION: on-demand vCPUs, Elastic IPs (5), VPCs (5), Lambda concurrency (1,000), etc. A DR environment that sits idle in a second Region often has DEFAULT quotas there — so when you fail over and try to scale to production size, launches fail against the limit. Design step: request quota increases in the DR Region ahead of time, matching production, and monitor quota usage with Service Quotas + CloudWatch alarms (and Trusted Advisor service-limit checks).

- Throttling: API Gateway (account and per-method/usage-plan limits), Lambda concurrency limits, DynamoDB capacity, and AWS API rate limits all return throttling errors; clients should implement exponential backoff with jitter. In a standby failover, verify these limits are raised in the DR Region too.

EXAM TIP: 'Failover to the secondary Region failed to scale' → service quotas were never raised there. 'Protect backend from being overwhelmed' → API Gateway throttling / usage plans.

3.3 NAT instance vs NAT Gateway (Domain 4, Task 4.4)

	NAT Gateway	NAT instance
Managed?	Fully managed, scales to 100 Gbps automatically, HA within its AZ	An EC2 instance you patch, size, and monitor; single point of failure unless you script failover
Cost	Hourly (~$0.045) + per-GB processed; can dominate small workloads' bills	Just the instance cost (t3/t4g.nano is a few dollars/month) — cheaper for low-traffic dev/test
Bandwidth	Up to 100 Gbps	Limited by instance type
Extras	Cannot be used as a bastion; no SGs (use NACLs)	Can double as a bastion host / port forwarding; has security groups; must disable source/destination check
Choose	Production, HA, high throughput, low ops	'Minimize cost for low-traffic non-production' or 'need bastion + NAT on one box'

- NAT Gateway placement cost trap: one shared NAT Gateway across AZs is cheapest in hours but incurs cross-AZ data charges and is a single point of failure; one per AZ costs more hours but avoids cross-AZ fees and survives AZ loss. Production answer = one per AZ; cost-only dev answer = one shared (or a NAT instance).

3.4 VPN bandwidth: single vs multiple tunnels (Task 4.4)

A Site-to-Site VPN connection gives two tunnels of ~1.25 Gbps each, but with a Virtual Private Gateway only one is active per connection (active/standby). To exceed 1.25 Gbps, terminate VPNs on a **Transit Gateway** and enable **ECMP** (equal-cost multi-path): multiple VPN connections/tunnels aggregate bandwidth (e.g., 4 connections ≈ 10 Gbps). Choosing Direct Connect speeds: pick hosted (50 Mbps–10 Gbps) for moderate steady loads, dedicated 1/10/100/400 Gbps for large; add a second connection or VPN for resilience.

EXAM TIP: 'VPN throughput is capped around 1.25 Gbps and must increase' → multiple VPN connections on Transit Gateway with ECMP. 'Consistent multi-Gbps, low latency' → Direct Connect.

3.5 Migrating applications into containers (Task 2.1)

Containerizing a legacy app: write a Dockerfile capturing runtime + dependencies; externalize configuration (environment variables, SSM Parameter Store, Secrets Manager) and state (S3, EFS for shared files, RDS/DynamoDB for data) so the container is disposable; push the image to ECR; run on ECS (Fargate for least ops, EC2 for cost/GPU) or EKS if the team

is Kubernetes-native; front with an ALB and scale on CPU/requests. **AWS App2Container (A2C)** automates discovering Java/.NET apps on servers, containerizing them, and generating ECS/EKS deployment artifacts. Docker Compose files can be deployed to ECS with minimal change.

EXAM TIP: 'Modernize a legacy Java/.NET app into containers with minimal manual effort' → App2Container → ECR → ECS/EKS. Stateful pieces must be externalized first — the same statelessness rule as horizontal scaling.

3.6 Secure access to ingestion points (Domain 3, Task 3.5)

Data lakes and streams need locked-down entry points: Kinesis and Firehose via IAM roles/policies and VPC interface endpoints (no public internet path); S3 ingest via bucket policies requiring TLS + specific principals, presigned URLs for external uploaders, or Transfer Family with SFTP keys; Lake Formation for column/row-level permissions once data lands; KMS encryption at rest across all of it; CloudTrail data events for audit of who read what.

4. Naming Updates & Out-of-Scope Services

4.1 Names that changed

Old name (you may see in courses)	Current name in the exam guide
Amazon QuickSight	Amazon Quick (same BI/dashboard service)
Amazon Kinesis Data Firehose	Amazon Data Firehose
Amazon Kinesis Data Analytics	Amazon Managed Service for Apache Flink
Amazon SageMaker	Amazon SageMaker AI
AWS Single Sign-On (SSO)	AWS IAM Identity Center
CloudFront Origin Access Identity (OAI)	Origin Access Control (OAC) — the recommended replacement
AWS CloudEndure / Server Migration Service	AWS Application Migration Service (MGN); DR variant = AWS Elastic Disaster Recovery
Snowmobile	Retired — multi-PB now via fleets of Snowball Edge devices

4.2 Officially OUT of scope — don't spend study time here

AWS lists these as out of scope. If any appear in an answer choice, they're distractors unless the question is explicitly about them (it won't be):

- **Developer tools:** CodeCommit, CodeBuild, CodeDeploy, CodeArtifact, CDK, CloudShell, Fault Injection Simulator, SDKs/Tools.
- **Compute/Database:** Lightsail, RDS on VMware.
- **ML:** Personalize, DevOps Guru, HealthLake, Elastic Inference, Inferentia, DLAMI/containers, MXNet/PyTorch/TensorFlow on AWS, DeepComposer.
- **Media:** all Elemental services (MediaConvert, MediaLive, MediaPackage, MediaTailor, MediaConnect), IVS.
- **Other:** all IoT services, MWAA (Airflow), Managed Blockchain, Sumerian, GameLift, Location Service, Cloud Map, Migration Evaluator, Distro for OpenTelemetry, Console Mobile App, Braket, Ground Station.

Earlier modules briefly mentioned Lightsail, CDK, Personalize, Forecast, Fraud Detector, Kendra, Bedrock, QLDB, and Timestream. Those are either out of scope or absent from the in-scope list; treat them as recognition-only and don't study them further.

5. Practice Questions on This Module

Practice Q1. A company's security team must let 300 remote employees reach private EC2 resources in a VPC from home laptops, authenticating with the corporate identity provider. Which service?

✗ Site-to-Site VPN — connects a network (customer gateway device), not individual laptops.

✗ Bastion host with SSH keys — unmanaged keys, no IdP integration, exposes port 22.

✓ AWS Client VPN endpoint with SAML federation to the IdP, associated with the VPC subnets. People-to-VPC = Client VPN.

Practice Q2. An organization wants developers in 30 accounts to launch only pre-approved, standardized environments (specific instance sizes, tagged, encrypted) without granting them permissions to create arbitrary resources.

✗ AWS Config rules — detect non-compliance after the fact; don't govern provisioning.

✗ Give developers CloudFormation + broad IAM — arbitrary resources become possible.

✓ AWS Service Catalog portfolios shared to the accounts, with launch constraints and a launch role that holds the permissions. Self-service provisioning of approved products = Service Catalog.

Practice Q3. A pilot-light DR design in a second Region failed during a test: the Auto Scaling group could not launch beyond 20 instances although production runs 80. Most likely cause and fix?

✗ Wrong instance type — would fail at instance 1, not 20.

✗ Route 53 misconfiguration — unrelated to launch failures.

✓ Default EC2 vCPU service quota in the DR Region; request quota increases matching production ahead of time and alarm on quota utilization. Quotas are per-Region and are a classic DR blind spot.

Practice Q4. A dev/test VPC with minimal outbound traffic pays more for its NAT Gateway than for its instances. Reduce cost with minimal risk to a non-production environment.

✗ Remove outbound access — breaks package installs.

✗ Add a NAT Gateway per AZ — increases hourly cost.

✓ Replace with a small NAT instance (t4g.nano) or a single shared NAT Gateway; for low-traffic non-production the instance is cheapest. The guide explicitly calls out NAT instance vs NAT Gateway cost.

Practice Q5. A platform team wants all accounts to launch workloads into centrally managed subnets in a shared networking account, so network controls live in one place. What enables this?

✗ VPC peering from each account — separate VPCs per account, not shared subnets.

✗ Transit Gateway — connects VPCs; doesn't let other accounts launch into your subnets.

✓ AWS Resource Access Manager to share the VPC subnets with the Organization (VPC sharing). Participants launch resources in the shared subnets; the owner controls routing, NACLs, and endpoints.

Practice Q6. Marketing needs Salesforce opportunity data landed in S3 nightly for analysis in Athena, with no custom integration code. Which service?

✗ Lambda calling the Salesforce API — custom code, auth handling, pagination, maintenance.

✗ DataSync — file/object transfer between storage systems, not SaaS APIs.

✓ Amazon AppFlow flow from Salesforce to S3 on a schedule, optionally with field mapping and Parquet output. SaaS → AWS with no code = AppFlow.

Coverage check — you're now aligned to the official guide

- All 14 task statements across the four domains: covered in Modules 1–4 and this supplement.
- All services on the official in-scope list: at least a one-liner recognition, with depth proportional to exam weight.
- Out-of-scope services identified so you don't waste study hours.

Final sequence: Modules 1–4 → this supplement → labs → timed practice exams, using the cheat sheet for daily recall drills.